

CS 601: CS Research Methodology Seminar

Chapter 1: Basics of Research

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What is research?

Novelty

Making contributions that were unknown previously. This means the work was not published in journals, conferences, or books.

Significance

The work must have high impact. It should be important, useful, or seminal, laying the foundation for future work.

Necessary component of research

- In natural science and engineering, “research” can be loosely defined as making *novel* and *significant* contributions to our understanding of the world, based on reproducible observations and verifiable results.
- In science, the emphasis is on discovering new theories, paradigms, approaches, algorithms, simulations, experiments, and so on.
- In engineering, the emphasis is to solve real-world problems with new technologies, designs, processes, methods, models, testing, and so on.
- **Is Computer Science a science discipline or an engineering discipline?**
- Computer Science overlaps with both Science and Engineering.
 - On the science side, CS has automata theory, algorithms, complexity analysis, formal verification, AI theories
 - On the engineering side, CS has programming, system development, testing, evaluation, AI models

Why do we do research?

- Let's ask a few students first ...
- To solve a problem
- To advance technology
- To benefit the society
- To answer a curiosity
- Just for the love of it!
- For money
 - Patents, licenses, start a company, external funding
- To generate publications for various reasons
 - Fame
 - Promotion
 - PhD degree
 - Passion

Different levels of research

- Solve a problem using existing methods. Write a documentation, a technical report, a manual, a README.txt. (**low innovation, low impact**)
- Improve an existing solution to an existing problem. Write a paper. (**low innovation, low-moderate impact**)
- Develop a new and much better solution to an existing problem. Write a paper. (**moderate-high innovation, moderate-high impact**)
- Identify a new problem. Generalize your solution so it can solve a class of problems. Write a paper. (**high innovation, high impact**)
- In CS, good research is usually characterized by “novel,” “original,” “creative,” “innovative,” “ground-breaking,” “transformative”, “significant,”

How to evaluate novelty?



Unpublished

Original work not found in existing academic databases or media.



Non-Obvious

It cannot be a small derivation or obvious improvement of previous work.



Creative

Introduces unconventional or ground-breaking thinking to a field.

Describing how to evaluate novelty

- What is not novel?
 - Something that is trivial, commonsense, already known in the public domain, already described in a published work,
- For something (e.g., method/algorithm) to be considered novel, it must be previously unknown or unproven, unpublished, the first of the kind, beyond what is known,
- In CS, research papers generally must have novelty
- This is not always true in other disciplines. In biology, medicine and physics, case studies, verification of methods, study groups, ... can be published as research papers.
- Then again they require a rigorous experimental standard.

How to measure significance of a research/paper?

- Below are some indicators:
 - Whether the work is published in a peer-reviewed, highly selective, prestigious conference or journal?
 - Whether the work is selected for awards such as best paper award, test-of-time best paper award, ACM best dissertation award,?
 - Whether the work received positive coverage in media?
 - How many times the work has been cited by others?
 - Google Scholar ([Link to scholar](#)) is a widely used resource for citation information of publications.
 - Each citation represents an impact or even a follow-up work.
 - Whether the work led to important applications and/or commercial product?
 - Whether the work laid the foundation for a new research area or discipline?

How to measure significance of a research or paper?

10 Yr

Most Significant Paper Award: Often given 10 years after publication to assess real long-term impact.

Informal Measures

- “ Citation counts from other researchers.
- 🌱 Used as a basis for other research programs.
- 👤 Reviewed and judged by experienced peers.



Understand Google Scholar

- Google Scholar crawls publications from conference proceedings and journals and analyzes citations from references.
- It also has PDF files of many papers. It is a very useful resource for finding publications.
- You can also find out the papers that referenced a paper.

The screenshot shows the Google Scholar interface with the search term "AI security". The results list several articles, each with a title, authors, publication details, and links to save, cite, or view the full text. The left sidebar contains filters for time, sorting, and type. The bottom section shows related searches and a list of papers that have cited the first result.

Google Scholar

AI security

About 3,970,000 results (0.09 sec)

Articles

Any time
Since 2025
Since 2024
Since 2021
Custom range...

Sort by relevance
Sort by date

Any type
Review articles

☐ include patents
☒ include citations

☐ Create alert

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The Researcher:

Myth: Researchers are old "nerds" or "geeks" who are unapproachable and incomprehensible.

Reality: Researchers are versatile, full of color, and normal people with hobbies.

Example: Richard Feynman played music, traveled, and wrote popular science books.



Who is not a Researcher?

Routine Roles

People who follow instructions like factory workers or accountants.

Sales & Ops

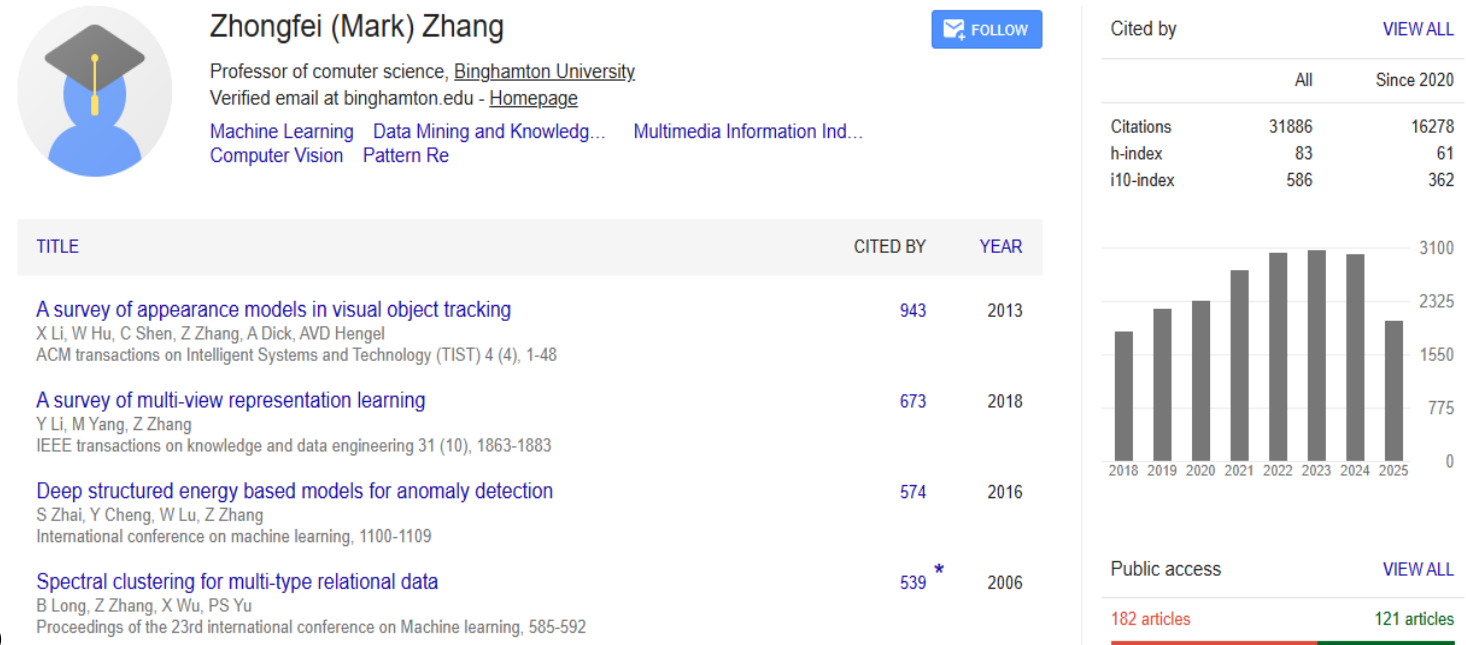
Professionals focused on commercial exchange rather than discovery.

Static Instructors

Teachers whose main job is delivery, not experimentation on new techniques.

How to measure the impact of a researcher?

- Researchers can create their profiles at Google Scholar. See an example below.
- Google Scholar uses three citation numbers to measure a researcher's overall impact.
 - Citations: Total citation count
 - h-index = h means there are h papers each received at least h citations
 - i10-index = i means there are i papers each received at least 10 citations.
- h-index was suggested in 2005 by Jorge E. Hirsch, a physicist at UC San Diego, as a tool for determining theoretical physicists' relative quality.



How to interpret h-index?

- h-index balances the number of publications and their citations.
- According to Hirsch, **for physicists**, a value for *h-index* of
 - about 12 might be typical for advancement to tenure (associate professor) at major US research universities
 - about 18 could mean a full professorship
 - 45 or higher could mean membership in the US National Academy of Sciences
 - after 20 years, a "successful scientist" would have an *h-index* of 20, an "outstanding scientist" would have an *h-index* of 40, and a "truly unique" individual would have an *h-index* of 60.
- Research.com keeps track of "Best Computer Science Scientists" ([scientists-rankings/computer-science](#)). To qualify, a researcher's h-index for computer science needs to be at least 30.
- Citation numbers (including h-index) for different disciplines/areas can vary significantly so use them with caution. Some citations may be generated by bad publicity.
- h-index, as well as most citation counts, do not consider the contribution of each author of a paper to the research.

What do Researchers do?

- The Ling & Yang book provides a list of 11 tasks that researchers often do in their day-to-day work.
- **Task 1.** Explore and come up with new ideas (i.e., novelty of research). To make sure that the idea is new, researchers must know well the frontier of their research areas by conducting an extensive literature search.
- **Task 2.** Validate and implement the new ideas to see if they work, and make them work well (i.e., significance of research).
- **Task 3.** Write up the research outcome in Tasks 1 and 2 as “manuscripts” or “papers,” and submit them to academic journals, conferences, book publishers, or other media outlets. Usually the research manuscript is first peer-reviewed by other researchers in the same field. Reviewers are always anonymous, sometimes reviews are **double-blind**. Many top journals and conferences may have an acceptance rate of only 20-30% for full-length papers.

What do Researchers do? (continued)

- **Task 4.** Review papers and manuscripts of other researchers to see if they are novel and significant enough to be accepted for publication.
- **Task 5.** Administrate academic journals and organize academic conferences, especially when one becomes an established researcher. These duties are often time-consuming, and are usually on a voluntary basis (i.e., no extra income), although in the long run, they help increase the academic and professional visibility of the researcher in their field.
- **Task 6.** Attend academic conferences to present papers, to learn the most recent research results, and to exchange research ideas and knowledge with other researchers. This is also an important “networking” time with other researchers in the same field. Established researchers may also be invited to give keynote speeches at conferences.

What do Researchers do?

- **Task 7.** Apply for research grants from government and other organizations to support research, including providing support to graduate students.
- **Task 8.** Supervise graduate students, especially Ph.D. candidates.
- **Task 9.** Teach graduate and undergraduate courses in universities and colleges.
- **Task 10.** Perform administrative work for the organizations that researchers work in. In the university this includes committee work in the department or college, write reference letters, and so on.
- **Task 11.** Commercialize research outcome (also known as Conducting “Technology Transfer”). Some research is applicable in the real world, and the results can be commercialized. In this case, researchers may also seek to protect their intellectual properties with patents and so on. A small number of researchers even create spin-off start-up companies to commercialize their research.

Skills & Abilities Important for Researchers

 **Passion & Focus:** Deep interest in solving complex problems over long periods.

 **Curiosity:** Thinking "out of the box" and questioning established views.

 **Integrity:** Scientific honesty and transparency in reporting results.

 **Quick Learning:** Strong analytical and problem-solving skills.

 **Diligence:** Willingness to spend months or years on breakthroughs.

 **Communication:** Being a "storyteller" who can convince peers.

Pros of being a researcher:

1. High career satisfaction:

Researchers have more freedom in choosing what interests them the most to work on than many other forms of employment. University professors also have flexible working hours in their university office, and have summer months in which they can focus on research, and travel for conferences and research visits. It is also highly rewarding when researchers make new discoveries and breakthroughs, which benefit people and are recognized by their peers.

Pros of being a researcher :

2. Protected research environment.

Researchers, especially university professors, have a relatively stable and protected position for research. These positions are usually not much affected by the short-term economic downturns and by the opinions of politicians. To protect academic freedom and give researchers time to produce high-impact results, most universities have a “tenure” system—tenured professors cannot be fired easily when they dissent from prevailing opinion, openly disagree with authorities of any sort, or if their research ‘quantity’ fluctuates due long-term research.

Pros of being a researcher :

3. **Reasonably good income.** Researchers usually have a Ph.D. degree, and their entry salary is often much higher than those with Master's or Bachelor's degrees. In addition, researchers' salary is also quite stable; there is usually no “bonus” or “penalty” if they accomplish a lot or a bit less for Tasks 1–10 in a particular year. This can be regarded as a disincentive, we suppose, as salespersons can earn more money if they sell more products. However, professors can earn an extra income if they hold secondary consulting or business positions, or if they are successful in technology transfer.

4. **Researchers are highly respected by most societies.** Nobel prize and awards for research accomplishments are always inspiring.

Cons of being a researcher:

1. Researchers often work more than 8 hours a day and 40 hours a week to stay competitive:

The amount of work needed for doing Tasks 1 to 11 well can be daunting and intimidating. However, most researchers are highly effective, and they learn to manage time well. In addition, research is a career, not just a “job.” Often researchers are intrigued by research questions, and enjoy thinking about them when they have extra time. They are also thrilled when a breakthrough is made.

Cons of being a researcher:

2. Long preparation time:

To become a researcher, one usually needs to study at a university for one's Bachelor's, Master's, and Ph.D. degrees. Sometimes they need to take a few years of post-doctoral positions after their Ph.D., before a more stable position is secured. However, Ph.D. candidates and post-doctoral fellows are partially researchers. They usually receive a small to moderate income during that time.

Cons of being a researcher:

3. Relatively narrow career choice.

Depending on the economy and available funding, in some years the number of job openings for post-doctoral fellows and professors in universities and research ers in companies can be limited. In addition, the number of Ph.D. students graduating in the last few decades is increasing. Thus, the competition for new positions for professors and researchers can be very high. Also, after you get a Ph.D. degree, you may be “overly qualified” for jobs for Master’s (or Bachelor’s) degrees.